AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

ANNUAL EXAMINATIONS (THEORY) 2023

Physics Paper II

Time: 1 hour 50 minutes Marks: 25

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are SEVEN questions. Answer ALL questions. Questions 6 & 7 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Draw a labelled diagram that demonstrates the passage of a light ray passing through a glass slab from air.

Space for diagram

Q.2. (Total 3 Marks)

a. State TWO conditions that must be fulfilled for a charge to experience magnetic force when it is placed in a magnetic field. (2 Marks)

b. State any ONE factor which causes the magnetic force to vary in terms of a charge particle. (1 Mark)

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Q.3. (Total 2 Marks)

Explain any TWO uses of electron gun as a source of electron beam.

Q.4. (Total 2 Marks)

Write any TWO uses of e-mails.

Q.5. (Total 3 Marks)

Is it possible that fusion between two small nuclei occur without a collision at extremely high energy? Justify your answer in THREE points.

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Q.6. (Total 6 Marks)

EITHER

a. Calculate the length of a simple pendulum if its frequency is 1.5 Hz. (6 Marks)

(**Note**: Take the acceleration due to gravity as 10 m/s^2.)

OR

b.

i. Define the terms, 'pitch' and 'loudness' of sound. (2 Marks)

ii. The intensity of a source of sound is 10^{-6} W/m^2 and the intensity of faintest audible sound is 10^{-12} W/m^2.

Calculate the sound level of the source. (4 Marks)

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Q.7. (Total 6 Marks)

EITHER

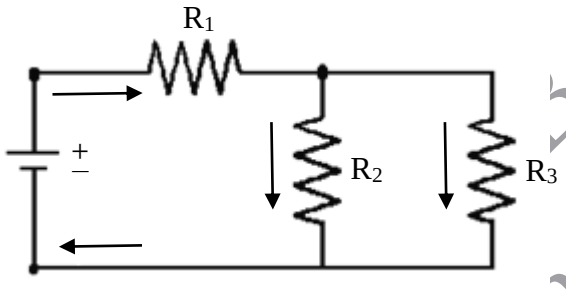
a. Calculate the equivalent capacitance of three capacitors 3 μF, 6 μF and 9 μF in a circuit if they are connected in

i. series combination. (3 Marks)

ii. parallel combination. (3 Marks)

OR

b. Calculate the total resistance in the given circuit, where $R_1 = 5\ \Omega$, $R_2 = 10\ \Omega$ and $R_3 = 15\ \Omega$. (6 Marks)



(**Note**: Please remember that R_2 and R_3 are in parallel combination and R_1 is in series combination of their product.)

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